Classifiers

Readings for today

- Chapter 4: Classification. James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). An introduction to statistical learning: with applications in R (Vol. 6). New York: Springer.

Topics

1. Logistic regression

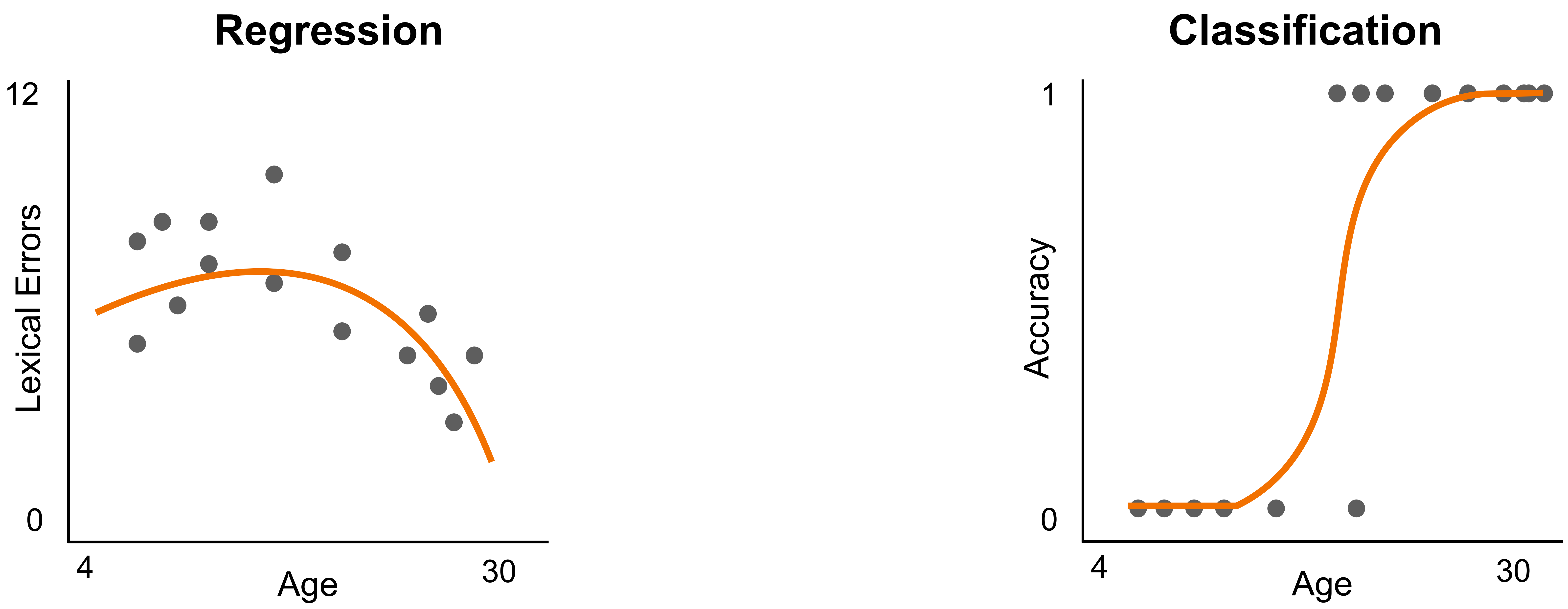
2. Threshold estimatation

3. Multi-class classification

Logistic regression

Classes of models

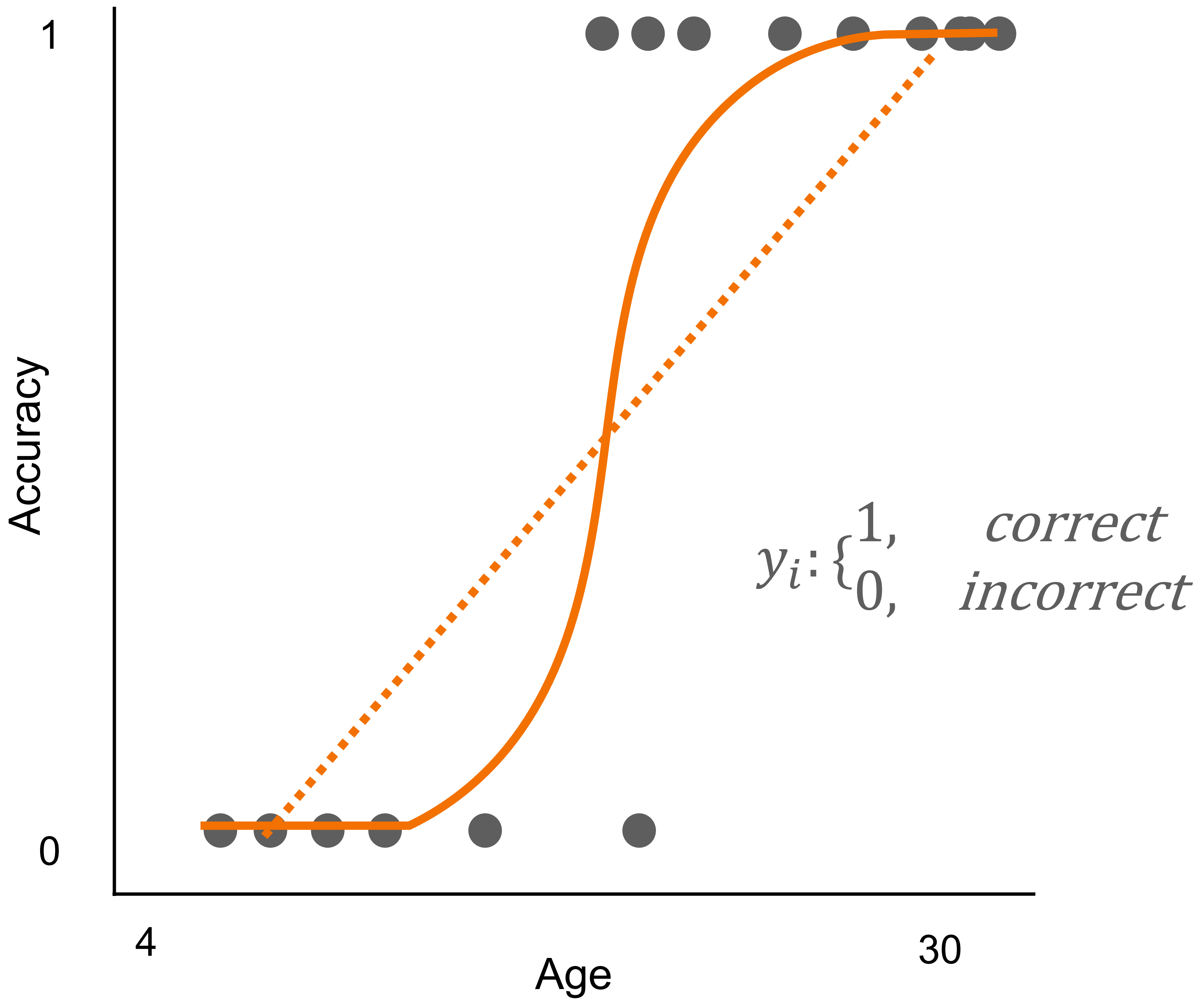
	Y Quantitative	Categorical
Regression	✓	
Classification		✓



(James et al. 2013)

Logistic regression

Q: what was the name of the social media platform that was getting outpaced by Facebook around 2008?



Linear regression

$$f(y_i|x_i,\beta_1,\beta_0,\sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-\sum_{i=1}^{n}(y_i-(\beta_0+\beta_1 x_i))^2}{2\sigma^2}}$$

Residual errors are normally distributed

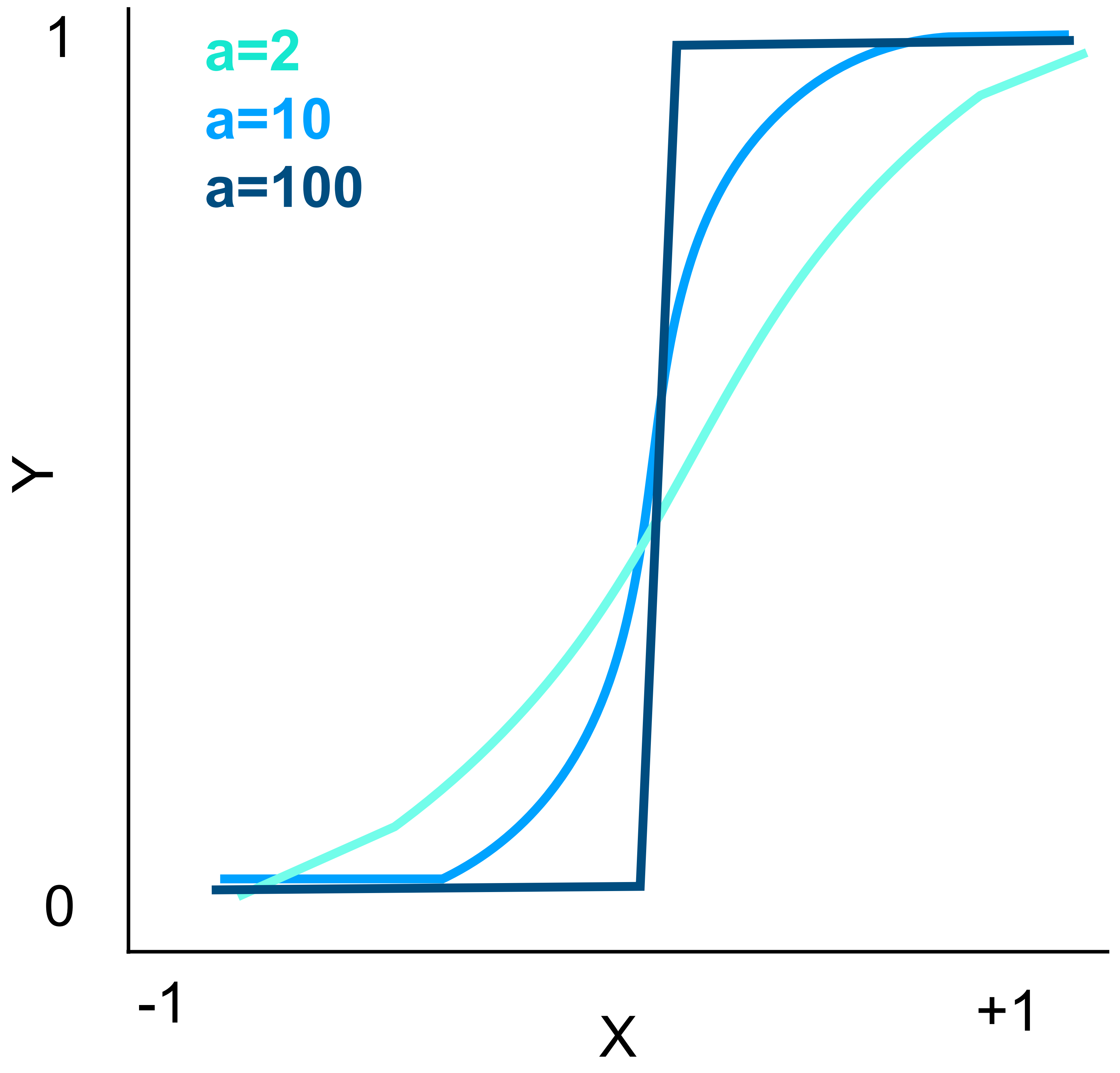
Logistic regression

$$p(y=1|x) = f(y_i|x_i,\beta_1,\beta_0,\sigma) = \frac{e^{\hat{\beta}_0+\hat{\beta}_1 X}}{1+e^{\hat{\beta}_0+\hat{\beta}_1 X}}$$

Residual errors are binomial (i.e., exist in one of 2 states)

The standard logistic function

Standard logistic function: $p(y=1|x) = \frac{e^{ax}}{1+e^{ax}}$



- Lower *growth rate* parameter (a), means slower transition between states of y (i.e., greater uncertainty).
- In a regression context better separation of y as x changes leads to larger a (i.e., $a = \sum_{j=1}^{p} \hat{\beta}_j X_j + \hat{\beta}_0$)

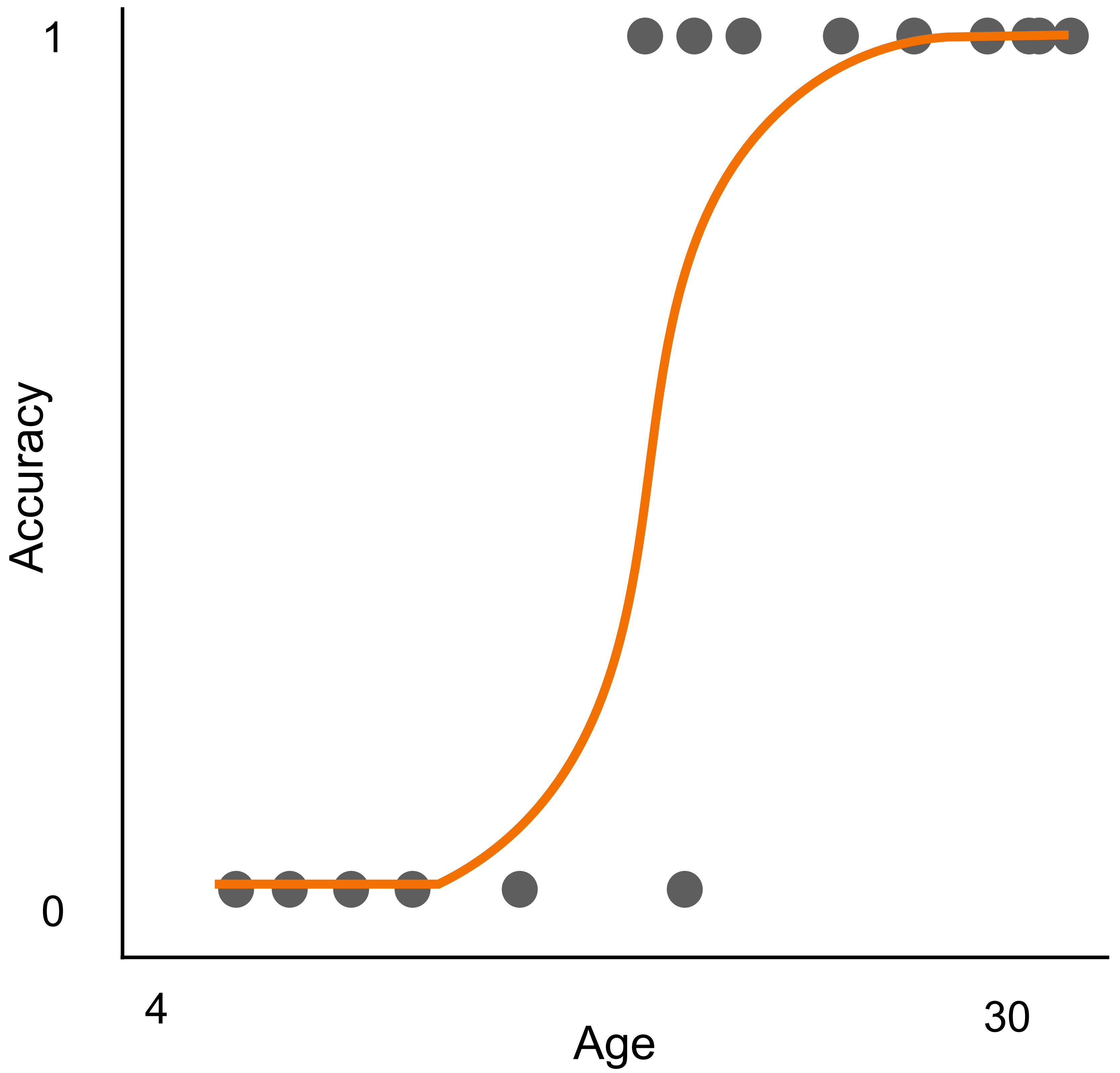
Odds

$$odds(y=1) = \frac{p(y=1)}{1+p(y=1)} = e^{ax}$$

$$ln(odds(y=1)) = ln(\frac{p(y=1)}{1+p(y=1)}) = ax = \hat{\beta}_0 + \sum_{j=1}^{p} \hat{\beta}_j X_j$$

(James et al. 2013)

Logistic regression



Logistic regression assumptions:

1. **Linearity in log odds**: predictors enter linearly in the parameters ($\eta = \log\left(\frac{p}{1-p}\right) = \beta_0 + X\beta$)
2. **Y is binary/dichotomous:** e.g., 0 or 1
3. **Enough observations** $(n \geq p)$: will run if $p \geq n$ but $\hat{\beta}\rightarrow\infty$
4. **Need adequate number of outcomes per event** for stable likelihood estimates
5. **No perfect multicollinearity:** No predictor is an exact linear combination of others.
6. **No complete separations:** X cannot perfectly predict Y otherwise $\hat{\beta}\rightarrow\infty$
7. **Independent observation** (no repeated measures, time series dependance, or there is only one value of y per observation).

(James et al. 2013)

Interpretation

Linear regression

$$\hat{y}_i = \beta_0 + \sum_{j=1}^{p} \beta_j x_{i,j} = 1 + 0.5x_i$$

Interpretation

For a 1 unit change in x, y changes 0.5, added to a baseline of 1.

Logistic regression

$\eta_i = \beta_0 + \sum_{j=1}^{p} \beta_j x_{i,j} = 1 + 0.5x_i$ = log odds that 'y=1'

$$p_i(y = 1) = \frac{e^{(\eta_i)}}{1 + e^{(\eta_i)}} = \frac{e^{(1+0.5x_i)}}{1 + e^{(1+0.5x_i)}}$$

For a 1 unit change in x, the log odds (aka "logit") of $y = 1$ changes by 0.5

When $X = 0$, the log-odds of '$Y = 1$' equal 1. This corresponds to odds of $e^1 = 2.72$, a probability of about 0.73 (73%).

(James et al. 2013)

Interpretation

Logistic regression

$$\eta_i = \beta_0 + \sum_{j=1}^{p} \beta_j x_{i,j}$$

Where is $\hat{y}$?

$$P = \frac{e^{(log\ odds)}}{1 + e^{(log\ odds)}} = \frac{e^{(\eta_i)}}{1 + e^{(\eta_i)}}$$

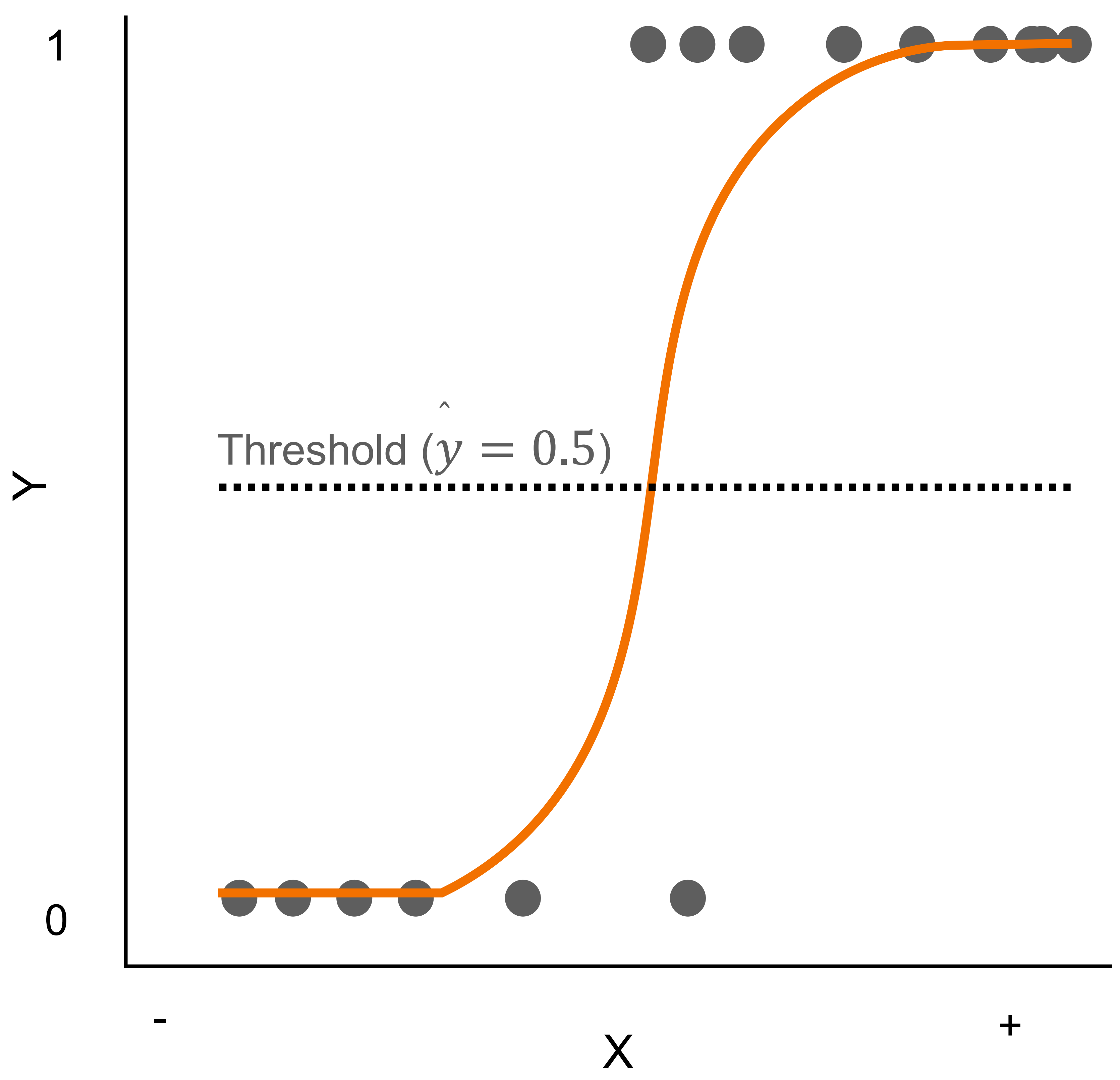
Log odds = 0: $\frac{e^{(0)}}{1+e^{(0)}} = \frac{1}{1+1} = 0.5\ (50\%\ probability\ y = 1)$ $\quad \beta_0 = 0$

Log odds = 1.386: $\frac{e^{(1.386)}}{1+e^{(1.386)}} = \frac{4}{1+4} = 0.8\ (80\%\ probability\ y = 1)$

Log odds = −1.386: $\frac{e^{(-1.386)}}{1+e^{(-1.386)}} = \frac{0.25}{1+0.25} = 0.2\ (20\%\ probability\ y = 1)$

(James et al. 2013)

Prediction



Estimating $\hat{y}_{\mathrm{i}}$:

- $\hat{f}(X)$ is still a continuous function.
- To predict $\hat{y}$, an *a priori* threshold needs to be determined to force a category for every $\hat{f}(x_i) = \hat{y}_i$.
- Position of this threshold determines the bias of your predictions.

(James et al. 2013)

Threshold Estimation

Cost of consequences drives threshold

	Actual 1	Actual 0
Predicted 1	True Positive (TP)	False Positive (FP)
Predicted 0	False Negative (FN)	True Negative (TN)

Case 1: Lower threshold
Missing a real case (false negative) is worse than a false alarm

- **Disease screening**
- **Fraud detection**
- **Suicide risk detection**

Case 2: Raise threshold
You only act when the model is very certain because false positives are worse.

- **Criminal conviction**
- **Expensive treatment**
- **Shutting down a factory line**

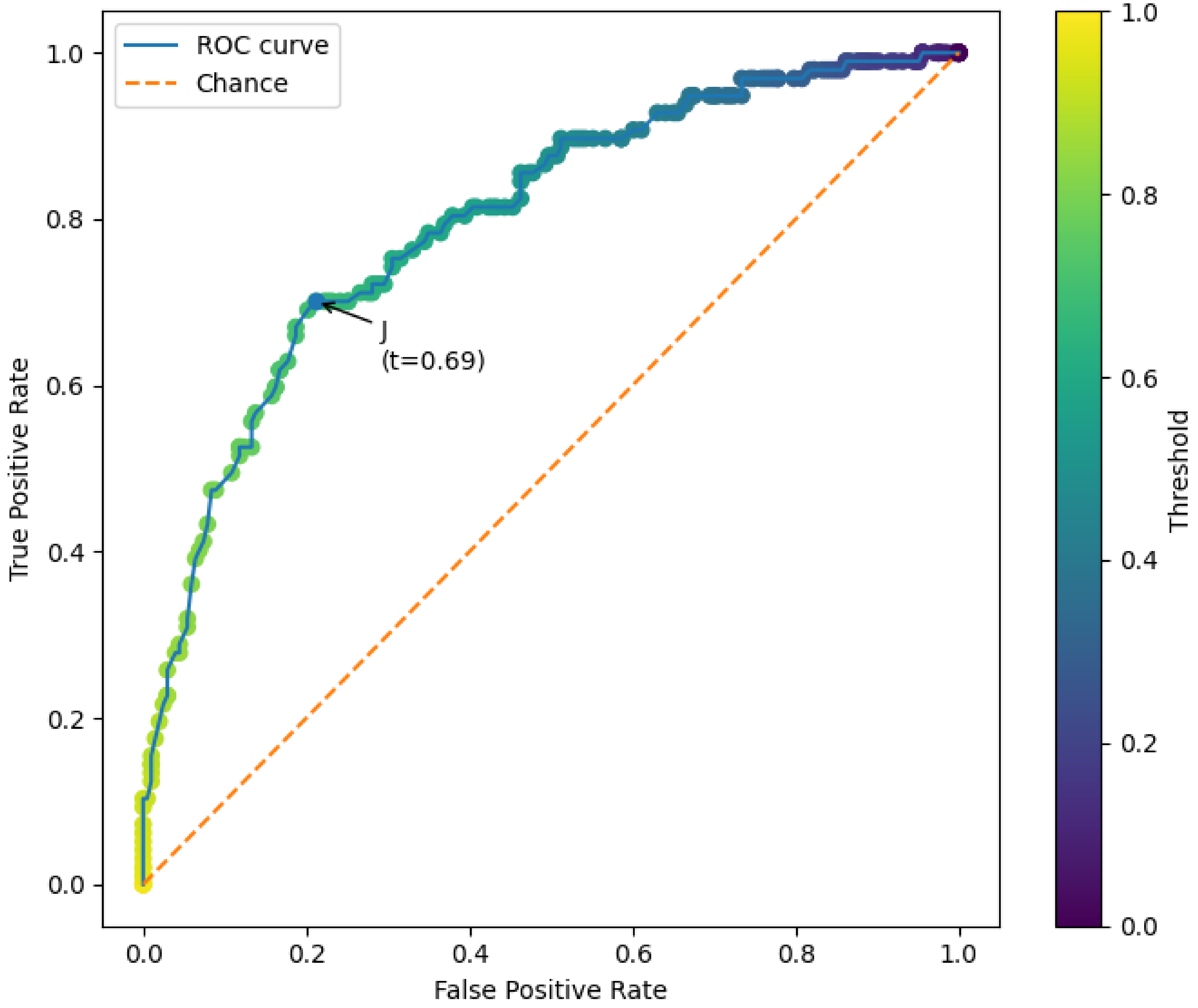
Data-driven thresholding: ROC Curve

	Actual 1	Actual 0
Predicted 1	True Positive (TP)	False Positive (FP)
Predicted 0	False Negative (FN)	True Negative (TN)

$\text{Sensitivity} = \frac{TP}{TP+FN}$ What proportion of real positives did we catch?

aka "***Recall***" OR *True Positive Rate (TPR)*

$\text{FalsePositiveRate(FPR)} = \frac{FP}{FP+TN}$ What proportion of real negatives did we catch?



$\text{J} = \text{Sensitivity} + \text{Specificity} - 1$

Largest J → maximizes correct detection while minimizing false alarms.

Data-driven thresholding: ROC Curve

$Q: Do\ people\ have\ the\ disease?$

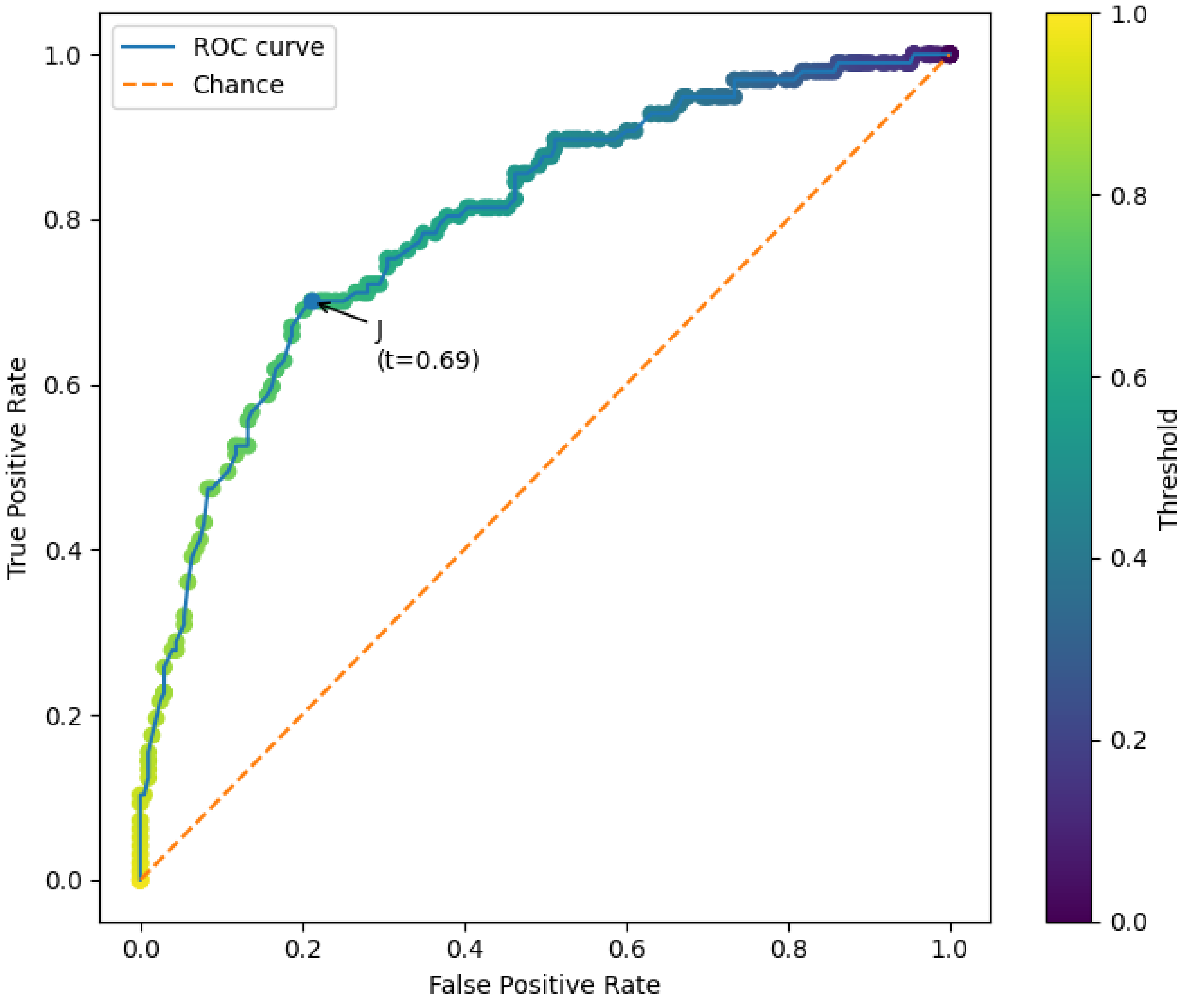
N = 10,000	**Actual 1**	**Actual 0**
Predicted 1	True Positive (80)	False Positive (120)
Predicted 0	False Negative (20)	True Negative (9780)

$$\text{Sensitivity} = \frac{80}{80+20} = 0.8$$

You identify 80% of people who actually have the disease!

$$\text{False Positive Rate} = \frac{120}{9780+120} \approx 0.012$$

You almost never (1.2%) said someone had the disease when they didn't have it!



Data-driven thresholding: Precision-Recall Curve

	Actual 1	Actual 0
Predicted 1	True Positive (TP)	False Positive (FP)
Predicted 0	False Negative (FN)	True Negative (TN)

$$\text{Recall} = \frac{TP}{TP + FN}$$

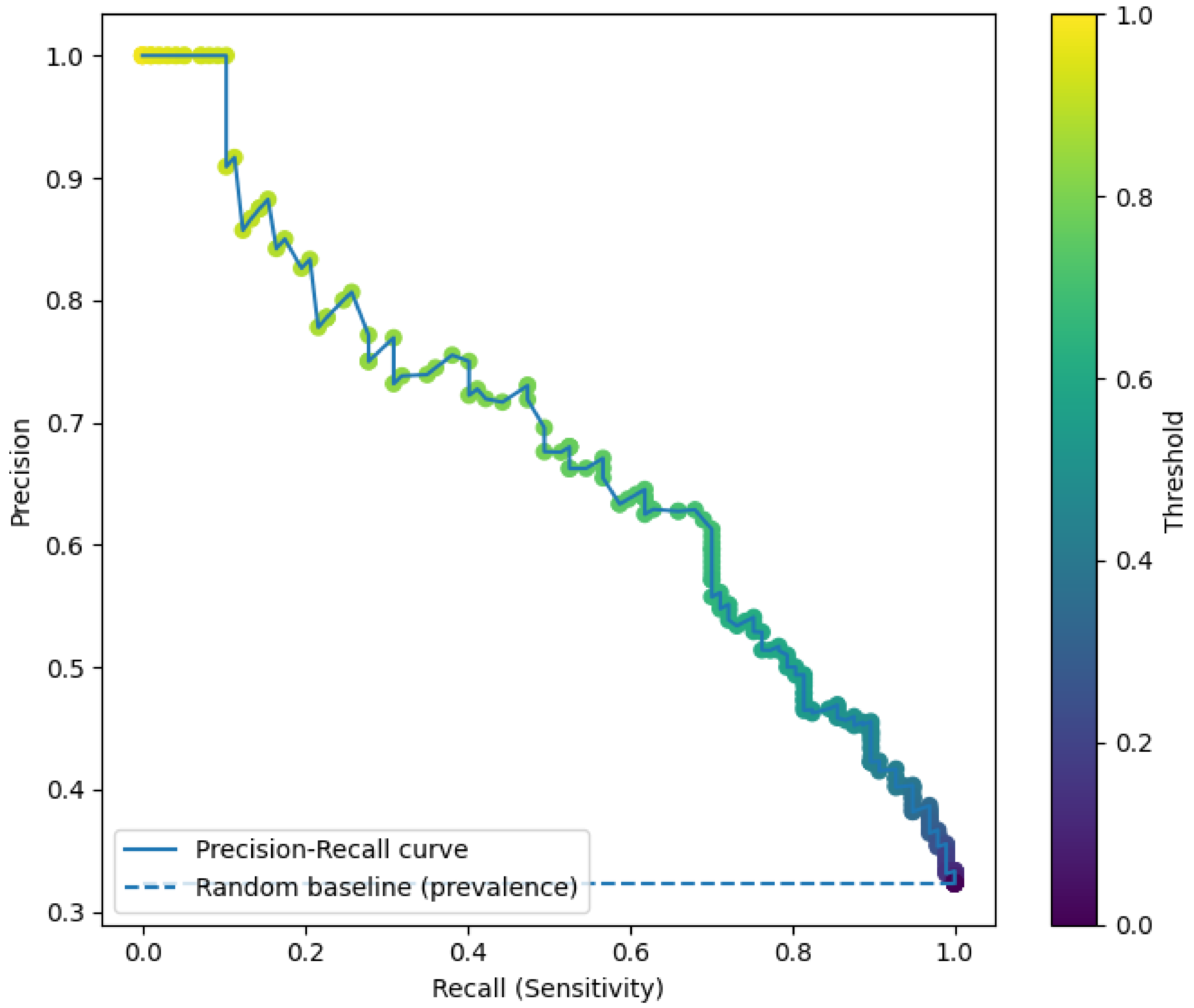
What proportion of real positives did we catch?

aka "***Sensitivity***" OR TPR

$$Precision = \frac{TP}{TP + FP}$$

When the models says the target is positive what proportion is it correct?

Precision − Recall Curve (PR)



Data-driven thresholding: Precision-Recall Curve

Q: *Do people have the disease?*

	Actual 1	Actual 0
Predicted 1	True Positive (80)	False Positive (120)
Predicted 0	False Negative (20)	True Negative (9780)

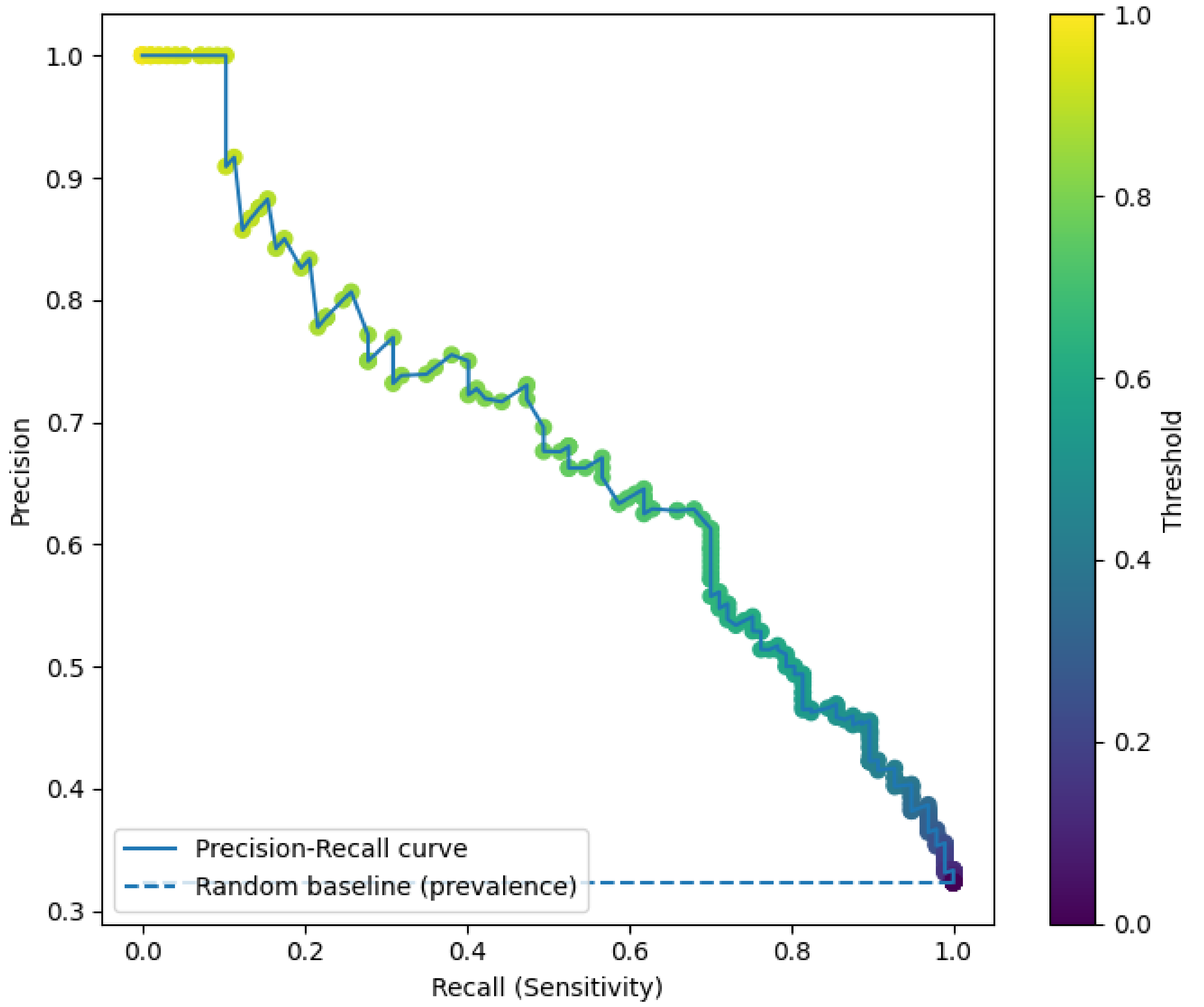
$$\text{Recall} = \frac{80}{80 + 20} = 0.80$$

You identify 80% of people who actually have the disease!

$$\mathit{Precision} = \frac{80}{80 + 120} = 0.4$$

Only 40% of the people you say have the disease actually have the disease.

Precision – Recall Curve (PR)



Combining detection & reliability: F1 score

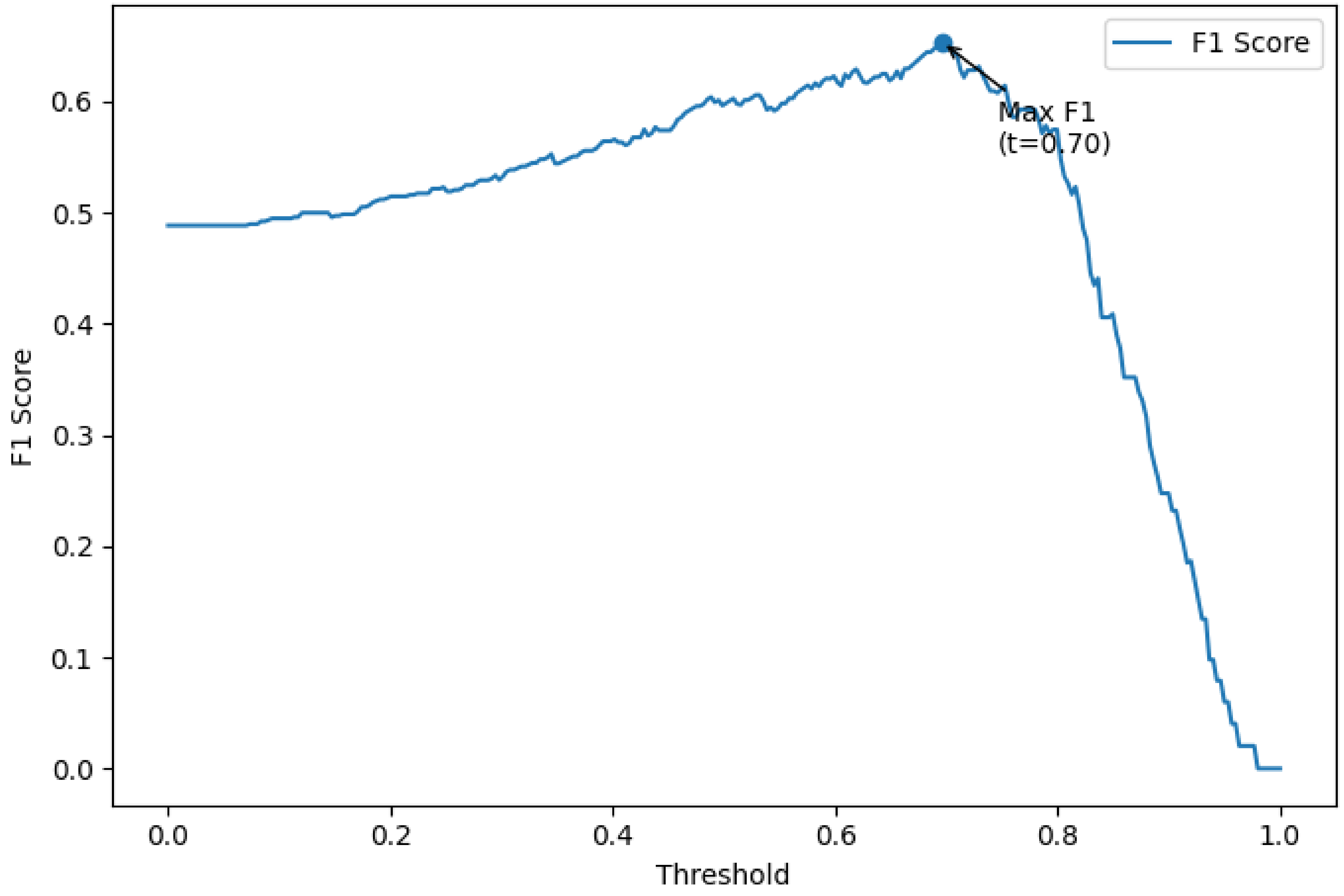
$$\text{Precision} = \frac{TP}{TP + FP} \qquad \text{Recall} = \frac{TP}{TP + FN}$$

$$F1\ score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

	What it optimizes	When to use
ROC / J	overall discrimination	balanced classes
Precision-Recall	reliability of positives	rare outcomes
F1	balances detection & reliability	cost is totally unknown

Precision: ‘If I call something positive, it must usually be true.’

Recall: ‘I want to catch as many true positive cases as possible’



(James et al. 2013)

Multiclass Classification

Multinomial logistic regression

Logistic regression

Possible states:
Accurate
Inaccurate

Latent quantity (log odds)

$$\eta_{accurate} = \log\left(\frac{P(Y = accurate)}{P(Y = inaccurate)}\right) = \beta_0 + \beta_1 X$$

Convert evidence to probability

$$P(y_i = accurate) = \frac{e^{(\eta_{accurate\,i})}}{1 + e^{(\eta_{accurate\,i})}}$$

Multinomial logistic regression

Possible states:
Treatment A
Treatment B
Control

Latent quantity (evidence score)

$$\eta_{A_i} = \beta_0 + \beta_{1A} X_i$$

$$\eta_{B_i} = \beta_0 + \beta_{1B} X_i$$

$$\eta_{Control_i} = 0$$

Softmax normalization (all probabilities sum to 1)

$$P(Y{=}k|X_i) = \frac{e^{\eta_{ki}}}{\sum_j e^{\eta_{ji}}}$$

Restating in Binary Logistic Regression terms

$$\eta_A = \log\left(\frac{P(Y = TreatmentA)}{P(Y = Control)}\right)$$

$$\eta_B = \log\left(\frac{P(Y = TreatmentB)}{P(Y = Control)}\right)$$

Multinomial logistic regression

Possible states:
Happy
Sad
Angry
Afraid

$$P(Y{=}k|X_i) = \frac{e^{\eta_{ki}}}{\sum_j e^{\eta_{ji}}}$$

$$P(Y{=}k|X_i) = \frac{e^{\eta_{ki}}}{e^{\eta_{Hi}} + e^{\eta_{Si}} + e^{\eta_{Angi}} + e^{\eta_{Afri}}}$$

$$P(Happy) + P(sad) + P(Angry) + P(Afraid) = 1$$

Let's do a test case!

$$\eta_{Happy_i} = \beta_0 + \beta_{1Happy}X_i$$

$$\eta_{Sad_i} = \beta_0 + \beta_{1Sad}X_i$$

$$\eta_{angry_i} = \beta_0 + \beta_{1angry}X_i$$

$$\eta_{afraid_i} = 0$$

$$P(Happy) = \frac{e^{\eta_{Hi}}}{e^{\eta_{Hi}} + e^{\eta_{Si}} + e^{\eta_{Angi}} + 1}$$

$$P(Sad) = \frac{e^{\eta_{Si}}}{e^{\eta_{Hi}} + e^{\eta_{Si}} + e^{\eta_{Angi}} + 1}$$

$$P(Angry) = \frac{e^{\eta_{Angi}}}{e^{\eta_{Hi}} + e^{\eta_{Si}} + e^{\eta_{Angi}} + 1}$$

$$P(Afraid) = \frac{1}{e^{\eta_{Hi}} + e^{\eta_{Si}} + e^{\eta_{Angi}} + 1}$$

Test case

Test data point: $X_i = 7$

1. Plug in to learned model

$$\eta_{Happy} = \beta_0 + \beta_{1Happy} \times 7$$

$$\eta_{Sad} = \beta_0 + \beta_{1Sad} \times 7$$

$$\eta_{angry} = \beta_0 + \beta_{1angry} \times 7$$

$$\eta_{afraid} = 0$$

2. Exponentiate to get 'unnormalized evidence'

$$e^{\eta_{Happy}}$$

$$e^{\eta_{Sad}}$$

$$e^{\eta_{Angry}}$$

$$e^{\eta_{Afraid}}$$

$$Z = e^{\eta_{Happy}} + e^{\eta_{Sad}} + e^{\eta_{Angry}} + e^{\eta_{Afraid}}$$

3. Compute ratio of class evidence to all evidence to obtain class probability

$$P(Happy \mid X = 7) = \frac{e^{\eta_{Happy}}}{Z} = 0.22$$

$$P(Sad \mid X = 7) = \frac{e^{\eta_{Sad}}}{Z} = 0.15$$

$$\boxed{P(Angry \mid X = 7) = \frac{e^{\eta_{Angry}}}{Z} = 0.55}$$

$$P(Afraid \mid X = 7) = \frac{e^{\eta_{Afraid}}}{Z} = 0.08$$

Take home message

- While logistic regression offers an intuitive extension to linear regression for classification problems, discriminant analyses are more flexible for classification in cases where your data consists of more than 2 categories.

- Ideally our estimates of cost will inform the threshold we specify, but if the cost is not known then choosing the threshold that allows for the greatest number of accurate True Positives and True Negative classifications is an option..

- Multinomial Logistic Regression Is a direct extension of the logic behind Binary Logistic regression to handle classification problems with more than 2 levels.